

PLEASE ANSWER ALL QUESTIONS IN EACH SECTION.

SECTION A: TRUE/FALSE QUESTIONS

[20 MARKS]

1. Questionnaire can be completed by many people at the same time. *True*
2. In the structured interview, the interviewees are allowed answer freely in their own *False* words.
3. Observation is part of noting and recognising people, objects and gets the information. *True*
4. Data sampling is important to speed up and improve effectiveness of data gathering. *True*
5. Below are examples of closed questions. *True*

Please select your age:

16-20

☐

21-25

☐

26-30

☐

Above 30

☐

6. Below are examples of closed questions. *False*

How long have you been in this position?

7. Below are examples of open-ended questions. **True**

How many hours do you spend on the system per week?

1-3 hours

☐

4-6 hours

☐

7-9 hours

☐

More than 10 hours

☐

8. This is an examples of open-ended questions. What are your key responsibilities? **True**
9. This is an examples of open-ended questions. How would you describe your **True** decision-making process?
10. Information along with financial, material and human resources is now regarded **True** as a key resource in many organizations.
11. A management information system departs from the traditional transaction **True** processing system in that it emphasizes the support of decision making in all its phases.
12. Knowledge work systems use common software tools to manipulate but not create new information. **False**
13. Computer supported collaborative work might include groupware for teams via networked computers. **True**
14. User involvement throughout the systems project is of little importance in the successful development of business information systems. **False**
15. Logical models show not only what a system is or does, but also how the system is **False**

implemented.

16. Process modelling is a technique for organizing and documenting the structure and flow of data through a system's processes and /or the logic, policies and procedures to be implemented by a system's processes. *True*
17. A data flow diagram (DFD) is a tool that depicts the flow of data through a system and the work or processing performed by that system. *True*
18. A data flow diagram is a process model. *True*
19. Data Flow Diagrams are a kind of flowchart. *False*
20. A process is work performed on, or in response to, incoming data flows or conditions. A synonym is transform. *True*

SECTION B: STRUCTURED QUESTIONS

[60 MARKS]

Question 1 (10 marks)

1. Questionnaires may seem to be a quick way to gather massive amounts of data which we can gather a lot of information through without spending time in face-to-face interviews. Explain TWO (2) appropriate conditions to use questionnaires. **(4 Marks)**
2. In addition to observing a decision maker's behavior, the systems analyst should observe the decision maker's surroundings for important clues as to how well the system fits the user. One method is Structured Observation of the Environment (STROBE). State TWO (2) elements that can be observed by applying STROBE. **(2 Marks)**
3. Joint Application Design (JAD) can be considered to cut both the time and cost of personal interviews. Briefly explain. **(2 Marks)**
4. Inductive organization of interview questions can be visualized as having **A** shape. Using this form, the interviewer begins with very detailed, often closed questions. The interview then expands the topics by allowing open-ended questions and more generalized responses. You should use **A** structure if you believe your interviewee needs to warm up to the topic.

In the **B** kind of structure, the interviewer takes a deductive approach by beginning with generalized, open-ended questions and then narrowing the possible responses by using closed questions. This interview structure can be thought of as **B** shaped.

Based on the above explanation, name the structure of A and B. **(2 marks)**

- 1) - Have a lot of respondent, easy to collect data from different people at the same time
- Limited time, faster than face to face interview.

- 2) 1. external information sources, get information from outside to support understanding and decision.
2. office lightning and color, to create a comfortable, productive and efficient environment.

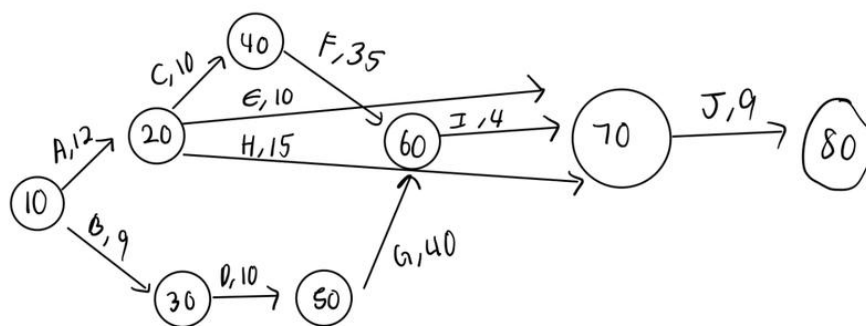
- 3) JAD involves user and manager to discuss system requirement together
This indicate reduce time since both of them can produce better idea and make a quick decision making.

- 4) A — Pyramid Structure, (close → open-ended question)
B — Funnel Structure, (open ended question → close)

Question 2 (10 Marks)

In the following table, you as a project manager for Smart Project in Company A have listed out all the activities (in weeks) for the project. Construct a network diagram by using a PERT diagram of the project and determine the earliest time and critical path of it.

Activity	Description	Predecessor	Expected Time
A	Select administrative staff	-	12
B	Survey	-	9
C	Select equipment	A	10
D	Prepare final plan	B	10
E	Interview applicants for staff	A	10
F	Purchase and take delivery of equipment	C	35
G	Construct the project	D	40
H	Development an information system	A	15
I	Install the equipment	F, G	4
J	Train staff	E, H, I	9



$$\text{Path 1: } A-E-J = 12+10+9=31$$

$$\text{Path 2: } A-H-J = 12+15+9=36$$

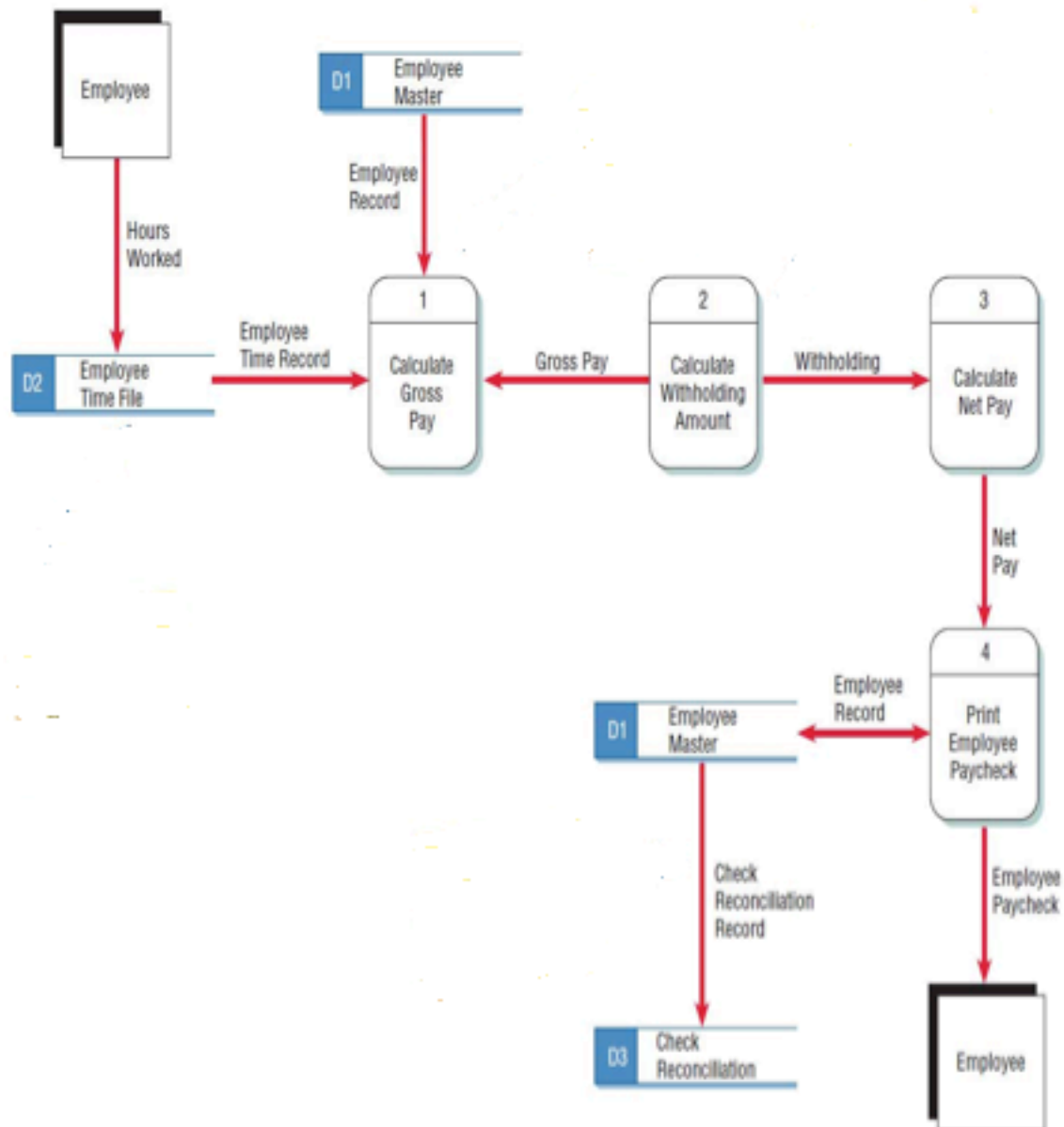
$$\text{Path 3: } A-C-F-I-J = 12+10+35+4+9=70$$

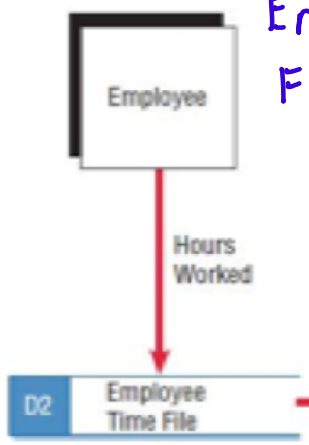
$$\text{Path 4: } B-D-G-I-J = 9+10+40+4+9=72$$

The earliest time is path 1, A-E-J while the critical path is path 4, B-D-G-I-J.

Question 3 (10 marks)

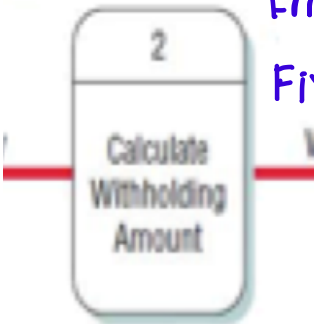
Several common errors made when drawing data flow diagram as below. Identify and explain the errors based on the listed numbers.





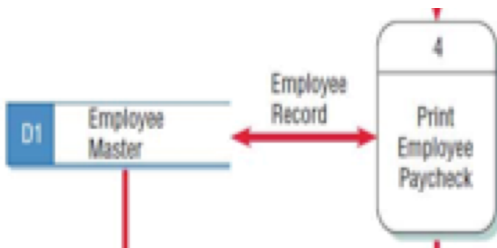
Error - Direct Entity to Data Store

Fix - Add a process between Entity and Data Store



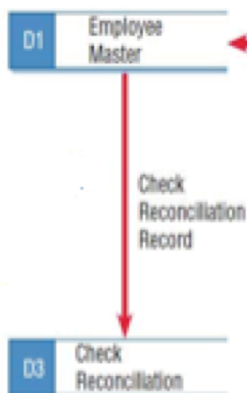
Error - not receive any input

Fix - Must have at least one input



Error - Data have two directions

Fix - Separate the data flow to two in order to indicate the different direction



Error - Direct data store to data store

Fix - Add a process between data store to data store

Question 4 (20 Marks)

OneMalaysia Communication Berhad is an established telecommunication company that has fourteen branches in Malaysia. Recently, the top management has decided to consider an information system called as HEROIN to help their growing business. You have been hired as the system analyst to handle the project. Before continue with the development of the project, you need to assess the economic feasibility from the budget that the company plans to invest. List below shows the information given by the company.

Development Costs

Hardware	RM 50,000
Software	RM 15,000
Training	RM 30,000
Consulting	RM 80,000
Data Conversion	RM 40,000

sensitivity factor = 1

Production Costs

Supplies	RM 15,000 per year
IS Salaries	RM 75,000 in year 1; 10% annual increase 0.1
Upgrades	RM 10,000 per year over 3 years

Benefits

Improve Customer Service	RM 150,000 in year 1; 20% increase annually 0.20
Increase productivity	RM 150,000 per year; 25% increase annually 0.25

Assumptions

Discount rate	35 % 0.35
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(a) Use the information above to create a cost-benefit analysis table (for 3 years) to assess the economic feasibility of the proposed system. **(15 marks)**

(b) Given the results of your cost-benefit analysis in (a) and the profitability index, do you recommend that the organization fund this development project? Why and why not? **(5 marks)**

Costs	Year 0	Year 1	Year 2	Year 3
Development Costs				
- Hardware	50 000			
- Software	15 000			
- Training	30 000			
- Consulting	60 000			
- Data conversion	40 000			
Total	215 000			
Production Costs				
- Supplies		15 000	15 000	15 000
- IS salaries		75 000	82 500	90 750
- Upgrade		10 000	10 000	10 000
Annual Production Costs		100 000	107 500	115 750
Present Value		74 074.07	58 984.91	47 045.67
Accumulated Costs		289 074.07	348 058.98	395 104.65
Benefits				
- Improve CS		150 000	160 000	216 000
- Increase productivity		150 000	187 500	234 375
Total		300 000	367 500	450 375
Present Value		222 222.22	201 646.09	183 051.36
Accumulated Benefits (Present Value)		222 222.22	423 868.31	606 919.67
Gain / Loss		(66 851.85)	75 809.35	211 815.02
Profitability Index	0.99			

entities : doctor, nurse, billing department

